

COURSE TITLE : COMPOUND DESIGN LAB
COURSE CODE : 3099
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/3
CREDITS : 3

TIME SCHEDULE

Module	Topics	Period
1	Develop gum rubber compounds and study the properties	16
2	Study the influence of fillers on physical properties of rubber	20
3	Design compounds with specific properties for different products	24
TOTAL		60

COURSE OUTCOME

Module	GO	Student will be able
1	1	To develop gum rubber compounds and study the properties
2	2	To study the influence of fillers on physical properties of rubber
3	3	To design compounds with specific properties for different products

**DETAILS OF LABORATORY WORK IN COMPOUND
DESIGN LAB**

1.1.0 Develop gum rubber compounds and study properties

- 1.1.1 Preparation of gum compounds with Natural Rubber and study the rheological properties and hardness
- 1.1.2 Preparation of gum compounds with Synthetic Rubber and study the rheological properties and hardness
- 1.1.3 Preparation of gum compounds with Reclaimed Rubber and study the rheological properties and hardness
- 1.1.4 Preparation of gum compounds with Blends of Rubber and study the rheological properties and hardness

2.1.0 Study the influence of fillers on physical properties of rubber compounds for desired

- 2.1.1 Study the influence of filler on hardness with different base polymer /blends (NR, SBR ,NR/HSR, NR/PBR)
- 2.1.2 Study the influence of filler loading (black) on hardness using NR,
- 2.1.3 Study the influence of filler loading (non black) on hardness using NR,
- 2.1.4 Study the effect of particle size of fillers on hardness
- 2.1.5 Study the influence of filler loading on abrasion resistance using NR,SBR, NR/HSR NR/PBR blends.
- 2.1.6 Study the influence of filler loading on tensile properties (TS, M300 EB) using NR ,SBR .NR/HSR, NR/PBR blends.
- 2.1.7 Study the influence of filler loading on tear resistance (TS,M300 ,EB) using NR ,SBR .NR/HSR, NR/PBR blends

3.1.0 Design compounds with specific properties for different products.

- 3.1.2 Design rubber compounds of 35, 45, 55, and 65 Shore A hardness using HAF,SAF, GPF and inspect test specimen.
- 3.1.2 Design compound having 35,45,55,65 shore A hardness with silica and inspect the test specimen
- 3.1.3 Design rubber compounds for pre determined specific gravity using black and non black fillers and inspect test specimen.
- 3.1.4 Study cure characteristics of rubber compounds using mooney viscometer, O D R.